

AWGN

Band-limited
AWGN

Parallel AWGN

Colored
Gaussian Noise

ECE253/CSE208 Introduction to Information Theory

Lecture 14: AWGN Channel

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- Chap 9 of *Elements of Information Theory (2nd Edition)* by Thomas Cover & Joy Thomas

White Gaussian Noise Channel (AWGN)

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A Gaussian channel is modeled as

$$Y_i = X_i + Z_i,$$

where X_i denotes the channel input and $Z_i \sim \mathcal{N}(0, N)$ is additive white Gaussian noise.

Theorem (AWGN channel capacity)

The channel capacity of a Gaussian channel with power constraint P and noise variance N is

$$C = \max_{f(x): \mathbb{E}(X^2) \leq P} I(X; Y) = \frac{1}{2} \log \left(1 + \frac{P}{N} \right) \text{ bits per transmission}$$

Proof of AWGN Channel Capacity

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$$I(X; Y) = h(Y) - h(Y|X) = h(Y) - h(Z) \quad (1)$$

$$\leq \frac{1}{2} \log[2\pi e \text{Var}(Y)] - \frac{1}{2} \log(2\pi e N) \quad (2)$$

$$\leq \frac{1}{2} \log[2\pi e(P + N)] - \frac{1}{2} \log(2\pi e N) \quad (3)$$

$$= \frac{1}{2} \log\left(1 + \frac{P}{N}\right), \quad (4)$$

where the maximum is attained when $X \sim \mathcal{N}(0, P)$. Thus, $Y \sim \mathcal{N}(0, P + N)$. Note that the channel capacity becomes infinite if we have zero-variance noise or unconstrained input.

Proof via Sphere Packing (1/2)

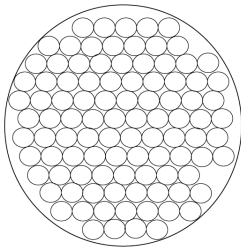
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The operational capacity (supremum of all achievable rates) of the Gaussian channel can be argued in a similar fashion as for discrete channels. The theorem can also be geometrically proved by sphere packing.



By LLN, as $n \rightarrow \infty$, we have the following

- Most received vectors $\{\mathbf{y}^n\}$ lie inside the n -dimensional sphere of radius $r_{\text{large}} = \sqrt{n(P+N)}$.
- A received vector $\mathbf{y}^n \sim \mathcal{N}(\mathbf{x}^n, N\mathbf{I})$ is contained in a sphere of radius $r_{\text{small}} = \sqrt{nN}$ around the sent codeword (with high probability).

Proof via Sphere Packing (2/2)

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- Decoding: assign everything in this sphere to the given sent codeword. An error happens only if $\mathbf{y}^n$ falls outside the sphere, which has low probability. Similarly, we can choose other codewords and their corresponding decoding spheres.
- Maximum number of non-overlapping spheres = Maximum number of codewords that can be reliably transmitted.

Q: How many non-overlapping spheres can be packed into the large sphere?

A: The n -dimensional volume of a sphere with radius r is $V_n(r) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)} r^n$

$$\Rightarrow 2^{nR} \leq \left(\frac{r_{\text{large}}}{r_{\text{small}}} \right)^n = \left(\frac{\sqrt{n(P+N)}}{\sqrt{nN}} \right)^n \Rightarrow R \leq \frac{1}{2} \log \left(1 + \frac{P}{N} \right).$$

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A more realistic channel model is the band-limited continuous AWGN

$Y(t) = (X(t) + Z(t)) * h(t)$, where $h(t)$ is the impulse response of an ideal bandpass filter with $f_{\max} = W$. Consider the channel being used over the interval $[0, T]$. We have

- $2W$ samples per second (sufficient for reconstruction due to the *Nyquist-Shannon sampling theorem*).
- Signal energy per sample is $PT/2WT$.
- Noise variance per sample is $N_0WT/2WT$ (double-sided noise power spectral density is $N_0/2$).
- Channel capacity per sample is $C = \frac{1}{2} \log(1 + \frac{P}{N_0W})$.

Therefore, the capacity of a band-limited AWGN is

$$C = W \log \left(1 + \frac{P}{N_0W} \right) \xrightarrow{W \rightarrow \infty} \frac{P}{N_0} \log_2 e \text{ bits per second}$$

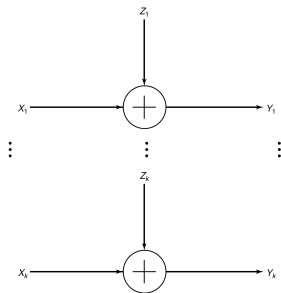
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We have a set of AWGN channels in parallel:

- channel output $Y_i = X_i + Z_i$, $i = 1, \dots, k$.
- independent noise $Z_i \sim \mathcal{N}(0, N_i)$.
- coupling power budget: $\mathbb{E} \left(\sum_{i=1}^k X_i^2 \right) \leq P$.
- mutual information:

$$\begin{aligned} I(X^k; Y^k) &= h(Y^k) - h(Y^k | X^k) = h(Y^k) - h(Z^k) \\ &\leq \sum_i h(Y_i) - h(Z_i) \\ &\leq \sum_i \frac{1}{2} \log \left(1 + \frac{P_i}{N_i} \right), \end{aligned}$$

where “=” is achieved by independent $X_i \sim \mathcal{N}(0, P_i)$.

Optimal Resource Allocation Problem

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The channel capacity can be obtained by solving the following optimization problem:

$$\max_{P_1, \dots, P_k \geq 0} \quad \sum_{i=1}^k \frac{1}{2} \log \left(1 + \frac{P_i}{N_i} \right) \quad (5a)$$

$$\text{s.t.} \quad \sum_{i=1}^k P_i = P \quad (5b)$$

Optimal Solution: Water-filling

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By Lagrangian relaxation, the optimal solution is obtained by the *water-filling* scheme: $P_i^* = (\nu - N_i)^+$, $i = 1, 2, \dots, k$, where ν satisfies $\sum_i (\nu - N_i)^+ = P$.

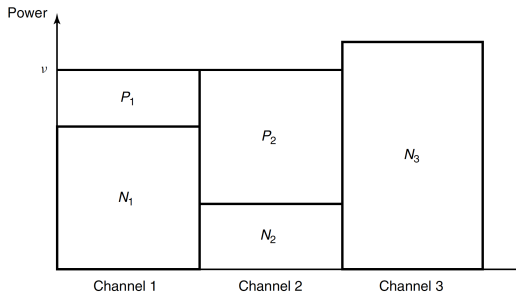


Figure: Water-filling solution: **allocate more power in less noisy channels**. $P_2 > P_1 > P_3 = 0$ and $P_1 + P_2 = P$. Channel 3 with the largest noise variance N_3 is the worst channel that gets no power allocation (i.e., no communication over this channel).

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- Since the noise and the input are independent, the covariance matrix of the output is $\mathbf{K}_Y = \mathbf{K}_X + \mathbf{K}_Z$.
- Maximize $I(X^n; Y^n) = h(Y^n) - h(Z^n)$ boils down to the following problem:

$$\max_{\text{Tr}(\mathbf{K}_X)=nP} \frac{1}{2} \log \left((2\pi e)^n |\mathbf{K}_X + \mathbf{K}_Z| \right).$$

- The optimal solution is obtained by *water-filling in the eigen-space* of $\mathbf{K}_Z$.
- Let $\mathbf{K}_Z = \mathbf{U} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{U}^\top$ denote its eigen-decomposition. Then,

$$\mathbf{K}_X^{\text{opt}} = \mathbf{U} \text{diag}(\tilde{\lambda}_1, \dots, \tilde{\lambda}_n) \mathbf{U}^\top,$$

where $\tilde{\lambda}_i = (\nu - \lambda_i)^+$ for $i = 1, \dots, n$ and $\sum_i \tilde{\lambda}_i = nP$ (to find the water level ν).

Gaussian Channels with Feedback

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- For a DMC, feedback does not increase the capacity, although it may help greatly in reducing the complexity of encoding and/or decoding.
- The same is true of memoryless AWGN channels. However, for channels with memory, where the noise is correlated across time, feedback does increase capacity.
- But, there is no simple explicit characterization of the capacity with feedback.

Capacity with feedback and the upper bounds.

$$C_{n,\text{FB}} = \max_{\text{Tr}(\mathbf{K}_X) \leq nP} \frac{1}{2n} \log \frac{|\mathbf{K}_{X+Z}|}{|\mathbf{K}_Z|} \leq \min \left\{ C_n + \frac{1}{2}, 2C_n \right\}.$$